HEMASHREE. P

Bachelor of Computer Science

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PROFESSIONAL SUMMARY

Final-year B.Sc. Computer Science student with strong interests in Machine Learning, Deep Learning, Computer Vision, and AI-driven scientific systems. Experienced in developing deep learning models for biomedical image classification using ResNet50, transfer learning, and Grad-CAM for explainable AI and cell image analysis. Skilled in Python, data preprocessing, model evaluation, and software development fundamentals including data structures and algorithms. Passionate about research-oriented AI applications in healthcare, image processing, and scientific analysis, with plans to pursue graduate studies in machine learning and computer vision.

TECHNICAL SKILLS

- **AI/ML & Computer Vision**: Machine Learning, Deep Learning, Convolutional Neural Networks (CNNs), Transfer Learning (ResNet50), Computer Vision, Biomedical Image Classification, Explainable AI (Grad-CAM), Image Processing, Data Preprocessing, Feature Extraction, Model Evaluation
- **Programming Languages**: Python, Java, C, R
- **Core Computer Science**: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering
- **Backend & Web Development**: Flask, API Integration, HTML, CSS, Basic JavaScript
- **Databases**: SQL, MySQL
- **Tools & Platforms**: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Kaggle
- **Currently Learning**: TensorFlow, Keras, PyTorch, Advanced Deep Learning, Representation Learning, Generative AI
- **Other Skills**: Debugging, Analytical Thinking, Problem Solving, Technical Documentation, Research Presentation, Team Collaboration

RESEARCH WORK

“Benchmarking Motion Blur Robustness in Modern Object Detectors: A Comparative Study of YOLOv8, RT-DETR, and GroundingDINO”

Independent Research | **2026**

- Conducted a comparative robustness study of YOLOv8, RT-DETR, and GroundingDINO under motion blur degradation.
- Developed an evaluation pipeline using Python, PyTorch, OpenCV, and modern object detection frameworks.
- Proposed and analysed Detection Retention Rate (DRR) and Confidence Retention Rate (CRR) metrics.
- Evaluated models on COCO validation images with synthetically generated motion blur.
- Authored a research manuscript and released the project repository; arXiv preprint submission planned.

“Deep Learning-Based Classification of Cell Images for Biomechanical State Analysis”

Published in *Proceedings of the International Conference on Sustainable Material (ICSM 2025)*
Tamil Nadu State Council for Science and Technology - July 2025

- Developed a Convolutional Neural Network (CNN) using transfer learning (ResNet50 architecture) for biomechanical state classification of cell images.
- Implemented Grad-CAM for model interpretability and feature visualization.
- Optimized pre-processing and model fine-tuning to improve classification performance.
- Contributed to experimental analysis and technical documentation for publication.

PROJECTS.

Leukemia Cell Classification Using ResNet50 and Grad-CAM

- Developed a deep learning-based leukemia cell image classification model using ResNet50 transfer learning
- Applied Grad-CAM visualization for explainable AI and model interpretability
- Performed image preprocessing, augmentation, model training, and evaluation
- Worked with biomedical image datasets using computer vision techniques
- Technologies: Python, TensorFlow/Keras, OpenCV, NumPy, Matplotlib, Scikit-learn

Kaggle Notebook - Chest X-Ray Medical Image Analysis (EDA)

- Created and documented an end-to-end machine learning workflow on Kaggle
- Included dataset preprocessing, visualization, model training, and evaluation
- Added markdown explanations and result analysis for better interpretability
- Demonstrated practical implementation of deep learning and computer vision concepts
- Used Python and data visualization libraries

OTSeg-inspired Multi-Prompt CLIP Prototype

- Developed a CLIP-based prototype inspired by the OTSeg paper for zero-shot semantic segmentation.
- Implemented multi-prompt evaluation to compare and rank text prompts against image embeddings using similarity scores.
- Explored vision-language alignment and foundational concepts of multimodal attention for segmentation-style tasks.

Food Waste Reduction Web Application

- A Flask-based web application connecting restaurants and NGOs to minimize food waste.
- Implemented secure user registration, donation tracking, and real-time update features.
- Designed backend database integration for efficient management of donation records.

INTERNSHIP

Web Development Intern - Velgrow Academy

- Developed a Python-based Nutrition Tracking Application using CustomTkinter.
- Implemented user authentication, nutrition logging, and diet recommendation features.
- Integrated backend logic with database management for efficient data storage and retrieval.
- Enhanced user experience through structured GUI design and responsive interface development.

ACHIEVEMENTS & AWARDS

- **Debugging Competition** - Recognized for optimizing complex code under time constraints January 2024
- **Debugging Competition** - Awarded for analytical problem-solving and programming precision January 2025
- **Paper Presentation** on Artificial Neural Networks - February 2026

CERTIFICATIONS

- **Python - Object-Oriented Programming (Infosys) -** Hands-on training on Python fundamentals and OOP concepts.
- **Frontend Development & Cloud Certification (IBM) -** Completed IBM-certified training in web development and cloud deployment.
- Smart Agriculture Monitoring System for Tamil Nadu Farmers - **Tamil Nadu Innovation Initiatives Hackathon, March 2026**

EDUCATION

Bachelor of Science in Computer Science (2023-2026) - Queen Mary's College, Chennai

VOLUNTEER EXPERIENCE

- **Community Service - Tree Planting Initiative:** Participated in local sustainability drives promoting environmental awareness.
- **Computer Knowledge Instructor -** Taught basic computer skills to school students.
- **College Volunteer (2023–2026) -** Actively contributed to organizing academic and extracurricular events. Collaborated with peers in coordinating college-level activities